AMRITA VISHWA VIDYAPEETHAM
SCHOOL OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

23CSE399 – Project Phase 1 (Technical Progress Report - Review 2)
March 28, 2026

Team Members		Guide	Project Type	Marks
CB.SC.U4CSE23547	SHEELA AKSHAR SAKHI	DR. G R RAMYA, DR. VANDHANA S Dept. of CSE	TYPE 1 RESEARCH	.../ 5
CB.SC.U4CSE23529	HASINI REDDY M			
CB.SC.U4CSE23761	KOUSIK SARMA LAKKARAJU			
CB.SC.U4CSE23363	HASWITHA K			
CB.SC.U4CSE23753	V. CHAKRAVARTHY			

1 Project Proposal and objectives

Overall Project Goal: Instead of merely implementing standard off-the-shelf algorithms, this project sets the objective to develop a novel extension to existing FedAvg and FedPer models. We will introduce the fully decentralized Federated Personalized Drift-Aware Attention Framework (FPDAF) specifically designed for multivariate healthcare time-series analysis (e.g., ICU risk forecasting).

Primary Novelty & Proposed Algorithmic Innovation: The definitive novelty of FPDAF lies in its planned hybrid integration. Unlike conventional FL architectures that maintain blind static global averaging, our extension proposes to intercept the FedAvg loop. We will inject a Novel CUSUM Extension directly into local active hospital nodes, enabling isolated Neural Network layers to act as an Automated Neural Trigger. This allows the model to "unfreeze" and adapt in real-time when patient physiology rapidly deteriorates, utilizing Temporal Attention to explain exactly which vitals caused the deviation.

Key Research Pillars:

- Novel FedAvg Extension: Uses Attention-Aware Weighting to prioritize higher-confidence hospital data during the global update instead of just simple patient-count averages.
- Novel CUSUM Extension: Repurposes statistical monitoring as an active neural manipulator for instant clinical adaptation.
- Novel Local Personalization: Extends standard FedPer nodes to focus on specific hospital demographics without losing shared global intelligence.

2 Introduction and Background

Clinical Context & Terminology:

- Sepsis: Life-threatening organ failure caused by infection; needs early detection.
- Deterioration: Rapid worsening of patient condition (vitals becoming abnormal).
- Profiling: Analyzing time-series data (Heart Rate, BP) to predict patient risk.

Continuous monitoring systems in Intensive Care Units (ICUs) collect immense streams of physiological data (HR, BP, SpO2). Early detection of patient deterioration using Deep Learning models

can prevent fatalities. However, massive clinical models require centralized data, violating privacy laws. While Federated Learning (e.g., FedAvg) resolves the data-sharing issue, it often fails to account for hospital-specific demographics and protocols (Accuracy Gap) or lacks clinician trust (Trust Gap).

Sustainable Development Goal (SDG) Mapping:

- Primary (SDG 3): Directly contributes to Good Health by enabling explainable patient risk prediction.
- Secondary (SDG 9 & 16): Supports scalable clinical innovation and ethical data privacy security.

3 Execution plan

The subdomains required involve: Data Engineering (custom ICU sequence windows), Deep Learning Base (extending LSTMs with novel Temporal Attention), Decentralization (Attention-Aware FedAvg), Customization (dynamic FedPer heads), and Continuous Adaptation (iterative feedback loops for real-time recalibration).

4 Fundamentals which you have learned in the semester

To prepare for implementation, the team completed an extensive curriculum:

- Textbooks & Papers: Thoroughly studied Federated Learning architectures (FedAvg, FedPer) to understand secure aggregation.
- PyTorch Engineering: Mastered custom training loops and gradient isolation techniques in PyTorch to separate "Global Backbones" from "Local Heads".
- The Privacy Gap: Studied international and local regulations—HIPAA and India's DPDPA 2023 (Section 8)—which mandate decentralized data handling.
- Statistical Mechanics: Learned the equations defining the CUSUM drift detection standard for monitoring temporal shifts in clinical residuals.

5 Architecture diagram and explanation of the design

The FPDAF architecture is a split-layer design as shown in Figure 1. It utilizes a centralized server for global weight aggregation while maintaining strictly isolated local personalization heads at each hospital node to ensure privacy and specialized clinical accuracy.

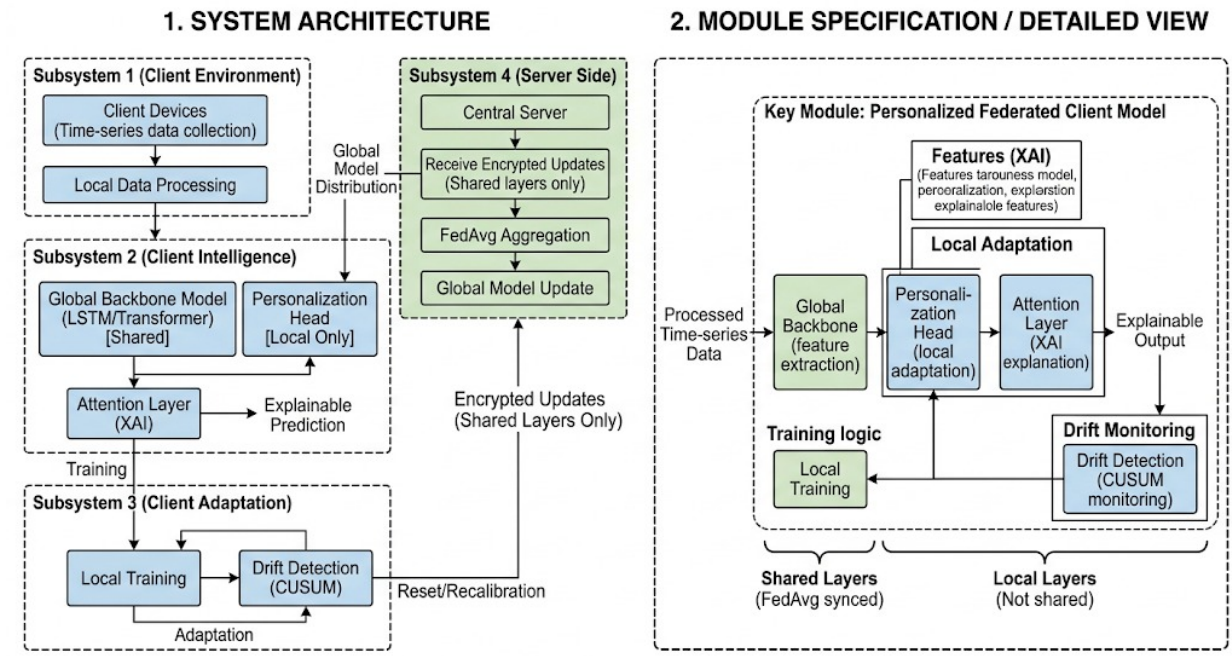

Figure 1: Master System Architecture: Highlighting the FPDAF extension for early ICU risk prediction.

The Attention Layer sits atop the backbone to compute temporal context vectors based on recent physiological spikes, giving transparent x_i timestamps feature importance. Finally, the Personalization Head is isolated completely to the hospital node; it never transfers its gradients to the server, ensuring the final clinical decision is strictly correlated to the local hospital's demographic distribution.

Comparative Framework Analysis:

Mechanism	Standard FedAvg	Standard FedPer	Proposed FPDAF
Aggregation	Full global averaging	Partial local-global split	Attention-Weighted Syncing
Drift Handling	Manual Retraining	Static Personalization	CUSUM Neural Trigger
Explainability	Black-box predictions	Local feature importance	Temporal Attention Weights
Personalization	None (One Global Model)	Multi-hospital site tuning	Dynamic Personalization

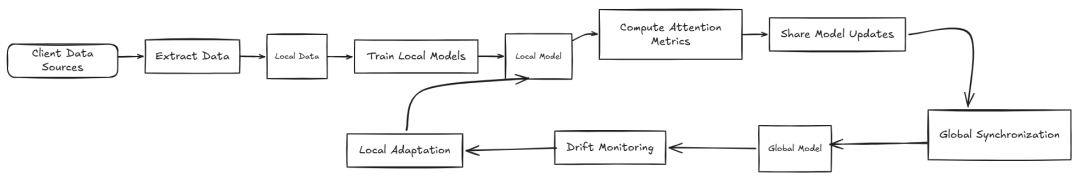

Figure 2: FPDAF System Dataflow Mapping - Illustrating end-to-end data processing.

6 Scientific Methodology & Neural Formalism

The innovation of FPDAF is formalized as a split-weight optimization problem across K hospital nodes. The local predictive output $\hat{y}_t^k$ is defined by:

$$\hat{y}_t^k = h_{\phi^k}\left(g_\theta(X_{1:t}^k) \cdot \alpha_t\right)$$

where g_θ is the Global Temporal Backbone (synchronized via FedAvg), h_{ϕ^k} is the Private Personalization Head (site-specific), and α_t represents the Temporal Attention context vector.

To handle temporal drift, we implement the CUSUM Neural Trigger. The system monitors the cumulative deviation of local prediction residuals:

$$S_t = \max(0, S_{t-1} + |y_t - \hat{y}_t| - \mu - k)$$

If $S_t > H$ (the statistical threshold), the framework automatically declares a patient state-drift, triggering a local backpropagation cycle on the personalization head ϕ^k while keeping θ frozen, preserving the global model's generalizability while adapting to the patient's acute deterioration.

7 Proposed Phase 2 & 3 Roadmap

The direct roadmap to full implementation is detailed in Table 2.

Target Area	Objective for Phase 2 & 3	Tentative Time
Data Preprocessing	Clean, Impute, and construct sequential windows for PhysioNet.	Ongoing – April 2026
Baseline Training	Code standard FedAvg on PyTorch locally for forecasting baselines.	April – June 2026
FPDAF Integration	Embed CUSUM triggers and Attention layers into personalized heads.	July – Sept 2026
Federated Simulation	Deploy 3-5 simulated Hospital "Clients" to test adaptation latency.	Oct – Dec 2026
Final Evaluation	Overall system performance benchmarking (Accuracy, F1-Score, AUC-ROC).	Jan – Feb 2027

Table 1: Project Execution Roadmap (2026-2027)

8 Summary of works completed in Phase 1

During this semester, the following milestones were successfully completed:

1. High-Era Dataset Aggregation: Secured a clinical stack including PhysioNet (2019, 2012, 2026), VitalDB, and EDT-Datasets.
2. Novelty Identification: Defined the FPDAF architectural extension fusing Federated Learning with proactive Drift Detection.
3. Mathematical Definition: Derived the formal split-weight optimization framework for localized personalization.
4. Data Engineering: Developed the initial Python sequence profiling script to handle ICU NaN issues and standardization.
5. Environment Setup: Established the official repository and verified standard FedAvg baseline feasibility.

Appendix: Data & Resource Access

Primary Project Repository: https://github.com/23CSE-TEAM-28/23CSE399-Project-Phase-1.git

Sepsis Challenge Data: https://physionet.org/content/challenge-2019/1.0.0/

PhysioNet 2012 Mortality: https://physionet.org/content/challenge-2012/1.0.0/

Trajectory Flow Matching (2026): https://physionet.org/content/clinical-trajectory-flow-icu/1.0.0/

VitalDB Dataset (2020): https://physionet.org/content/vitaldb/1.0.0/

EDT-Datasets (2021): https://github.com/mommand/EDT-Datasets

Full Literature Review (25 Papers): https://docs.google.com/spreadsheets/d/1G_zAbr6MI_bYcCHEXfkWjhmvKzBEDKr0/edit#gid=417550448

References

[1] E. Churaev, I. Kisel, and A. Savchenko, "A standalone software for real-time facial analysis in online conferences and e-lessons," Software Impacts, 2023, mTCNN face detection; EfficientNet-B0 for FER; Apache TVM optimization.

[2] R. Jothimurugesan, P. S. Hiremath, and S. Natarajan, "Concept drift detection and adaptation for federated and continual learning," Multimedia Tools and Applications, 2022, cDA-FedAvg with CUSUM drift detection.

[3] A. Arivazhagan, V. Aggarwal, A. Bajpai, and S. Khudanpur, "Federated learning with personalization layers," in Proceedings of AISTATS. Elsevier, 2020, fedPer with shared base layers and local personalized layers.

[4] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in Proceedings of AISTATS. PMLR, 2017, fedAvg using local SGD and global model averaging.

[5] Y. Liu, X. Chen, Y. Liu, and Z. Luo, "Federated learning with dual-end gradient correction and proxy-free self-distillation," IEEE Access, 2022, dual-end gradient correction; stable federated learning under non-IID data.

[6] X. Yang, J. Jung, J. Lu, K.-W. Lim, and L. Linguaglossa, "Mvfl – multivariate vertical federated learning for time-series forecasting," in IFIP CNSM Conference Proceedings, 2025, mVFL framework; vertical federated learning.

[7] R. Kalakoti, S. Nomm, and H. Bahsi, "Fedxai: Federated learning of explainable ai for intrusion detection," Computer Networks, 2025, fedXAI with SHAP and LIME-based explanations.

[8] M. Noseda, A. D. Luca, L. V. Briel, and N. Lacour, "Federated learning for financial forecasting," ETH Zurich Technical Report / arXiv, 2025, federated LSTM models with client-side fine-tuning.

[9] A. Karamanlıoğlu, B. Demirel, O. Tural, O. T. Doğan, and F. N. Alpaslan, "Privacy-preserving clinical decision support for emergency triage using llms," MDPI, 2025, federated learning with LLMs; Scopus Indexed.

[10] R. Wang, L. Bai, R. Rayhana, and Z. Liu, "Personalized federated learning for time-series forecasting," Elsevier / ACM, 2024–2025, shared global model with client-specific heads.

[11] R. Abdel-Sater and A. B. Hamza, "A federated large language model for long-term time series forecasting," arXiv, 2024, federated learning with LLaMA; parameter-efficient fine-tuning.

[12] M. Maher, O. F. Oun, M. S. Mesmeh, and R. ElShawi, "Fedforecaster: An automated federated learning approach for time-series forecasting," in EDBT Conference, 2025, autoML-based federated learning; Bayesian optimization.

[13] S. Gupta and V. Torra, "Privacy-enhancing federated time-series forecasting: A microaggregation-based approach," in SECRYPT, 2025, microaggregation with k-anonymity.

[14] W. Yuan, G. Ye, X. Zhao, Q. V. H. Nguyen, Y. Cao, and H. Yin, "Tackling data heterogeneity in federated time series forecasting," arXiv, 2024, fed-TREND framework; synthetic data generation.

[15] D. Tang, N. Yang, Y. Li, Z. Zhu, Z. Jin, and D. Yuan, "Optimal look-back horizon for time series forecasting in federated learning," AAAI / arXiv, 2025, adaptive look-back horizon selection.

[16] Y. Zhang, J. Wang, H. Liu, and X. Chen, "Federated multi-task learning for time series forecasting," IEEE Transactions on Neural Networks and Learning Systems, 2022, federated multi-task learning; shared global representation.

[17] G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, "Interpretable multivariate time-series forecasting with attention mechanisms," in NeurIPS, 2021, attention-based encoder-decoder; temporal and variable-level attention.

[18] G. Amato, C. Gennaro, A. Matteucci, and L. Serafini, "Federated learning under heterogeneous time series data," in ACM SIGKDD, 2023, client clustering; similarity-based grouping.

[19] G. L. Rocca, A. S. Podda, R. Martoglia, and M. Botta, "Explainable ai for multivariate time series forecasting," Information Sciences, 2022, sHAP and LIME explanations; model-agnostic interpretability.

[20] M. Bernardi, P. Giordani, and F. Ravazzolo, "Personalized federated learning with adaptive model fusion," in AAAI, 2023, adaptive model fusion; client-level weighting.

[21] C. Düsing and P. Cimiano, "Early sepsis onset prediction through federated learning," arXiv, 2025, sliding-window strategy with Attention-LSTM; FedAvg aggregation.

[22] A. Qayyum, K. Ahmad, M. A. Ahsan, A. Al-Fuqaha, and J. Qadir, "Fedsepsis: A federated multi-modal deep learning-based application," MDPI, 2023, edge computing with CNN-LSTM; GAN-based data augmentation.

[23] J. Lei, J. Zhai, Y. Zhang, J. Qi, and C. Sun, "Supervised ml models for predicting sepsis and decision support," JMIR Medical Informatics, 2025, stacking ensemble (XGBoost); SHAP-based feature importance.

[24] W. Pan, Z. Xu, S. Rajendran, and F. Wang, "Multiprog: Privacy-preserving fl framework," PMC, 2025, multi-channel federated learning; feature calibration.

[25] A. Maurya, R. Haripriya, M. Pandey, J. Choudhary, and D. P. Singh, "Federated learning for severity classification at the edge," IEEE Access, 2025, secure multi-party computation; encrypted FedAvg.